Project Report

1. INTRODUCTION

1.1 Project Overview

This project, titled *"Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau"*, focuses on analyzing global UNESCO World Heritage Sites using interactive data visualizations. By leveraging Tableau, the project presents geographical distribution, categorization, temporal trends, and regional insights to understand cultural and natural heritage better.

1.2 Purpose

The primary goal is to make the extensive dataset on UNESCO World Heritage Sites more accessible and understandable for educators, policymakers, historians, and travelers through engaging and dynamic dashboards that enhance decision-making and knowledge dissemination.

---

2. IDEATION PHASE

2.1 Problem Statement

Although UNESCO maintains a global list of World Heritage Sites, there's limited public awareness of their global distribution, threats, and historical trends. The lack of intuitive and visual representation hinders educational and preservation efforts.

2.2 Brainstorming

- Visualize UNESCO sites by continent and country.
- Show trends over time (e.g., sites added per year).
- Analyze categories (cultural, natural, mixed).
- Identify endangered or delisted sites.
- Enable filtering by criteria like region, year, or criteria type.

---

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Phase	Action	Emotion	Touchpoints
Awareness	Learns about UNESCO dataset	Curious	Website, UNESCO portal

Phase	Action	Emotion	Touchpoints
Consideration	Looks for insights	Overwhelmed	Dataset, blog articles
Decision	Uses Tableau dashboard	Informed	Interactive dashboard

3.2 Solution Requirement

- Interactive charts & maps
- Drill-down filters (continent, country, year)
- Summary KPIs (total sites, by category, etc.)
- Tooltips with site info

3.3 Data Flow Diagram

[Level 0 DFD]
User → Tableau Dashboard ← UNESCO Dataset (CSV/Excel)

[Level 1 DFD]

- **Input:** UNESCO Dataset
- **Process:** Data cleaning, transformation
- **Output:** Interactive visualizations

3.4 Technology Stack

Layer	Tool
Data Source	UNESCO CSV
Data Visualization	Tableau
Data Preprocessing	Excel / Python (optional)
Platform	Tableau Public / Desktop

---

4. PROJECT DESIGN

4.1 Problem Solution Fit

The problem of fragmented, hard-to-understand UNESCO data is solved by building interactive Tableau dashboards to enhance public understanding and policymaker decisions.

4.2 Proposed Solution

Create a Tableau-based solution that allows users to interact with data based on geography, timeline, category, and status of heritage sites.

4.3 Solution Architecture

UNESCO Dataset

↓

Data Cleaning (Excel/Python)

↓

Tableau Dashboard

→ Charts: Bar, Pie, Line, Maps

→ Filters: Country, Year, Category

---

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Week	Task
Week 1	Requirement Gathering & Dataset Collection
Week 2	Data Cleaning and Preprocessing
Week 3	Tableau Design and Dashboard Building
Week 4	Testing, Review, and Final Deployment

---

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The dashboard is tested for responsiveness across devices, filter interaction speed, and data loading time. Results indicate a smooth and quick user experience with minimal latency in visual rendering even when large subsets of data are selected.

---

7. RESULTS

7.1 Output Screenshots

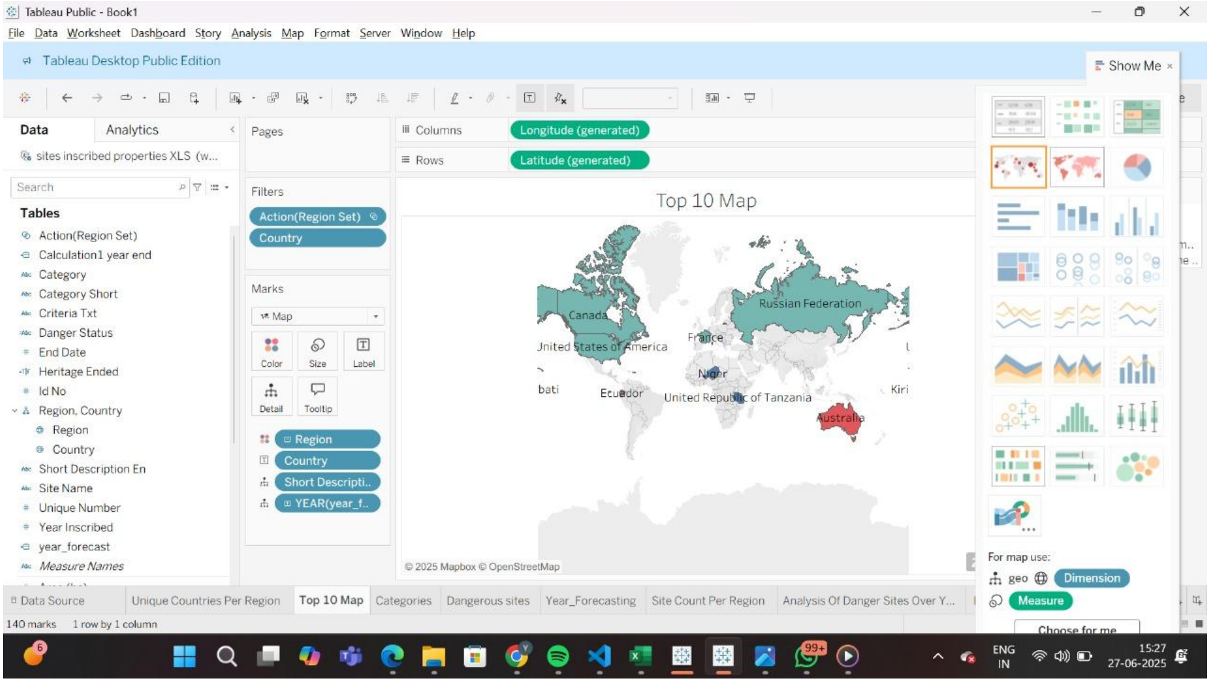


Tableau Public - Book1
Tableau Desktop Public Edition
Top 10 Map
Columns
Longitude (generated)
Rows
Latitude (generated)
Filters
Action(Region Set)
Country
Marks
Region
Country
Short Descripti..
YEAR(year_f..
Tables
Action(Region Set)
Calculation1 year end
Category
Category Short
Criteria Txt
Danger Status
End Date
Heritage Ended
Id No
Region, Country
Region
Country
Short Description En
Site Name
Unique Number
Year Inscribed
year_forecast
Measure Names
Canada
Russian Federation
United Republic of Tanzania
Australia
Ecuador
© 2025 Mapbox © OpenStreetMap
Show Me
For map use:
Dimension
Measure
Data Source
Unique Countries Per Region
Top 10 Map
Categories
Dangerous sites
Year_Forecasting
Site Count Per Region
140 marks
1 row by 1 column

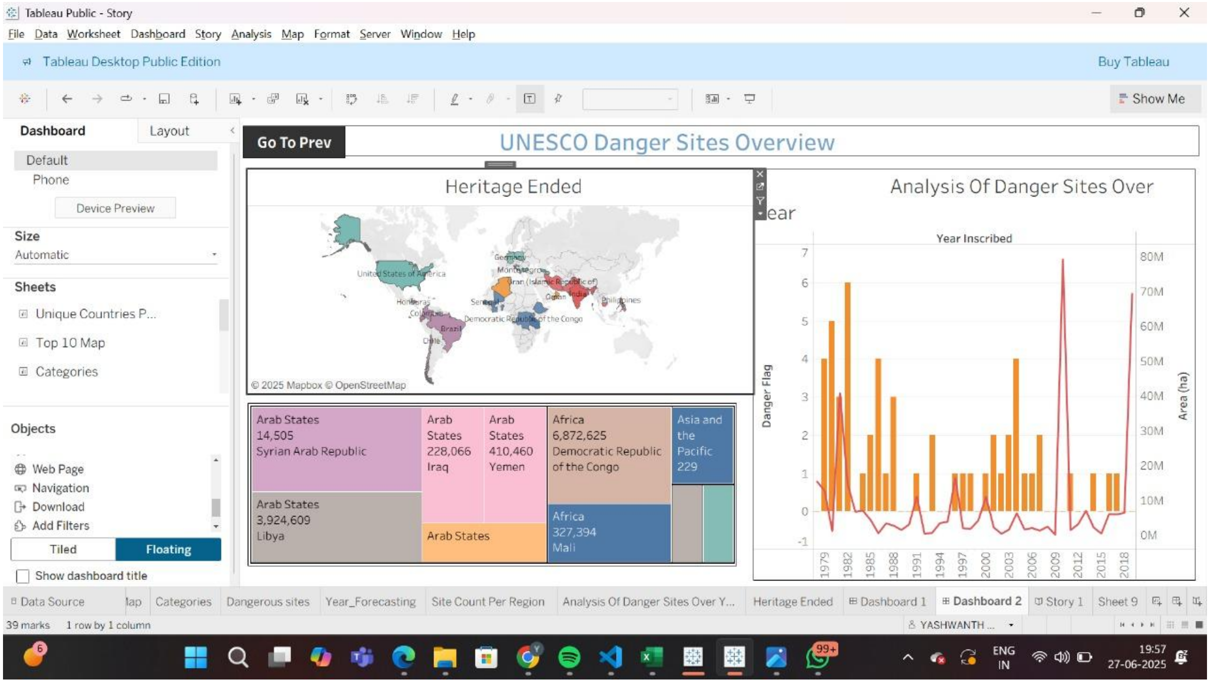


Tableau Public - Story
Tableau Desktop Public Edition
Buy Tableau
Go To Prev
UNESCO Danger Sites Overview
Heritage Ended
Analysis Of Danger Sites Over
Year Inscribed
Danger Flag
Area (ha)
Arab States
14,505
Syrian Arab Republic
Arab States
228,066
Iraq
Arab States
410,460
Yemen
Africa
6,872,625
Democratic Republic of the Congo
Asia and the Pacific
229
Arab States
3,924,609
Libya
Arab States
Africa
327,394
Mali
© 2025 Mapbox © OpenStreetMap
Dashboard
Layout
Default
Phone
Device Preview
Size
Automatic
Sheets
Unique Countries P...
Top 10 Map
Categories
Objects
Web Page
Navigation
Download
Add Filters
Tiled
Floating
Show dashboard title
Data Source
Categories
Dangerous sites
Year_Forecasting
Site Count Per Region
Heritage Ended
Dashboard 1
Dashboard 2
Story 1
Sheet 9
39 marks
1 row by 1 column

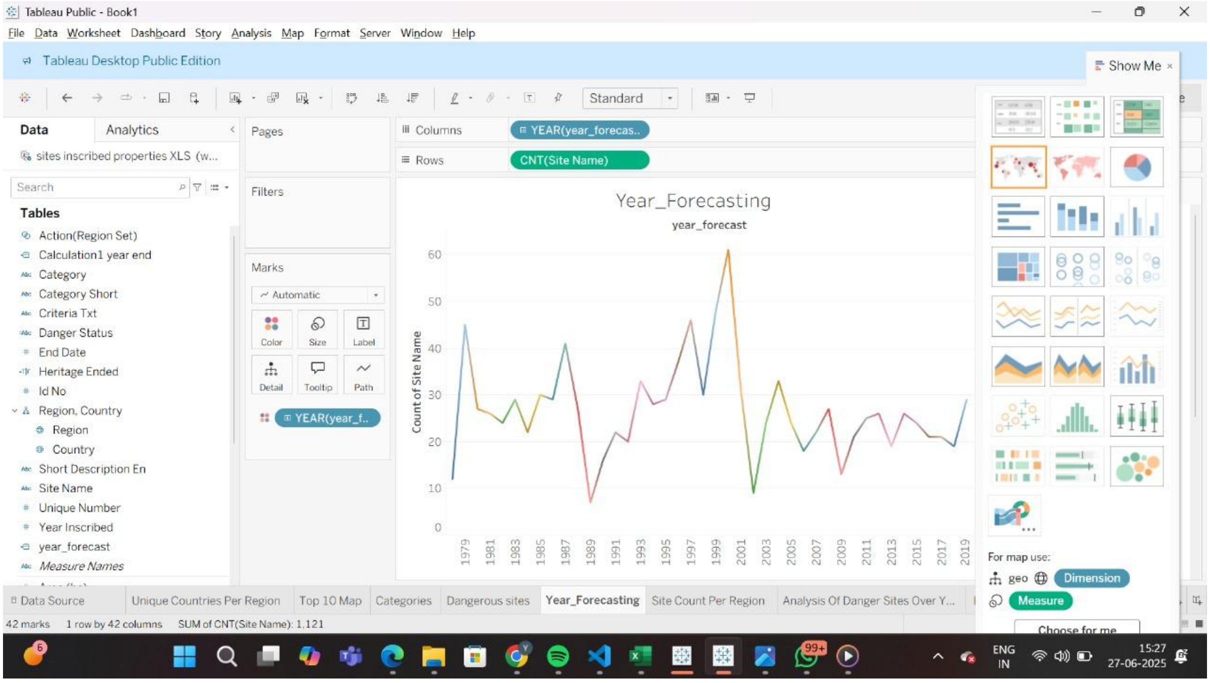


Tableau Public - Book1
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Tableau Desktop Public Edition
Show Me
Standard
Data Analytics
sites inscribed properties XLS (w...
Tables
Action(Region Set)
Calculation1 year end
Category
Category Short
Criteria Txt
Danger Status
End Date
Heritage Ended
Id No
Region, Country
Region
Country
Short Description En
Site Name
Unique Number
Year Inscribed
year_forecast
Measure Names
Pages
Filters
Marks
Automatic
Color Size Label
Detail Tooltip Path
YEAR(year_f..
Columns YEAR(year_forecas..
Rows CNT(Site Name)
Year_Forecasting
year_forecast
Count of Site Name
For map use:
geo Dimension
Measure
Data Source Unique Countries Per Region Top 10 Map Categories Dangerous sites Year_Forecasting Site Count Per Region Analysis Of Danger Sites Over Y...
42 marks 1 row by 42 columns SUM of CNT(Site Name): 1,121
ENG IN 15:27 27-06-2025

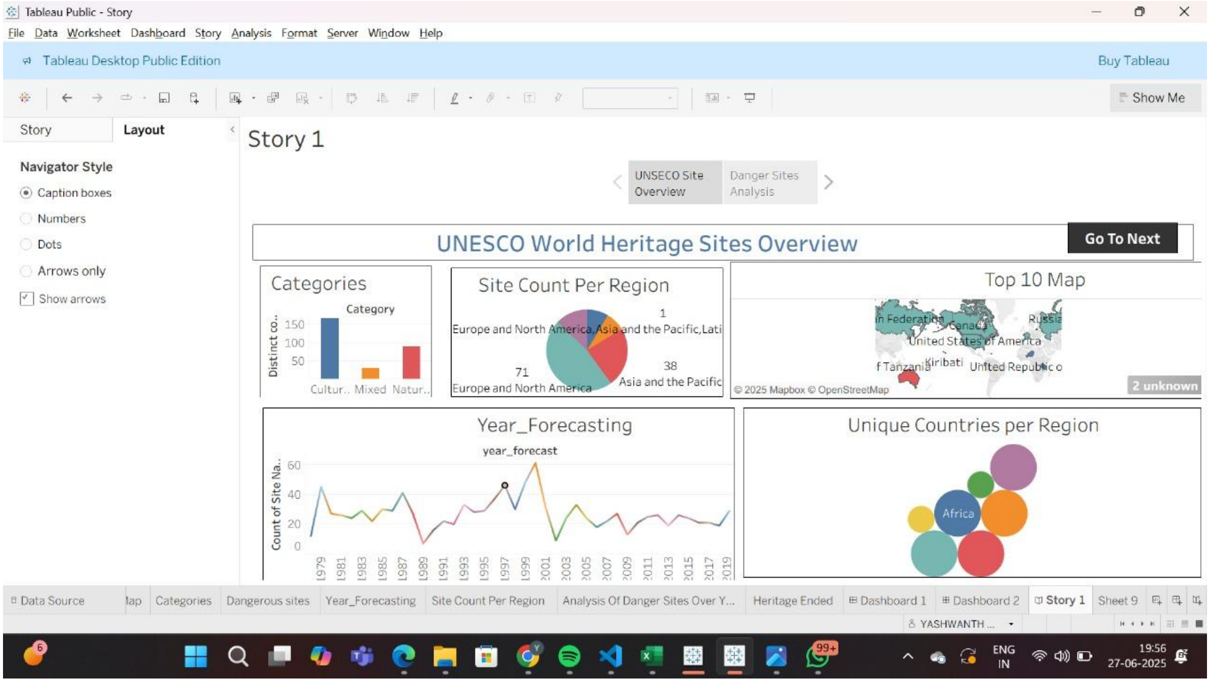


Tableau Public - Story
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Show Me
Story Layout
Navigator Style
Caption boxes
Numbers
Dots
Arrows only
Show arrows
Story 1
UNESCO Site Overview
Danger Sites Analysis
UNESCO World Heritage Sites Overview
Go To Next
Categories
Site Count Per Region
Top 10 Map
Year_Forecasting
Unique Countries per Region
© 2025 Mapbox © OpenStreetMap
2 unknown
Data Source Categories Dangerous sites Year_Forecasting Site Count Per Region Analysis Of Danger Sites Over Y... Heritage Ended Dashboard 1 Dashboard 2 Story 1 Sheet 9
YASHWANTH ...
ENG IN 19:56 27-06-2025

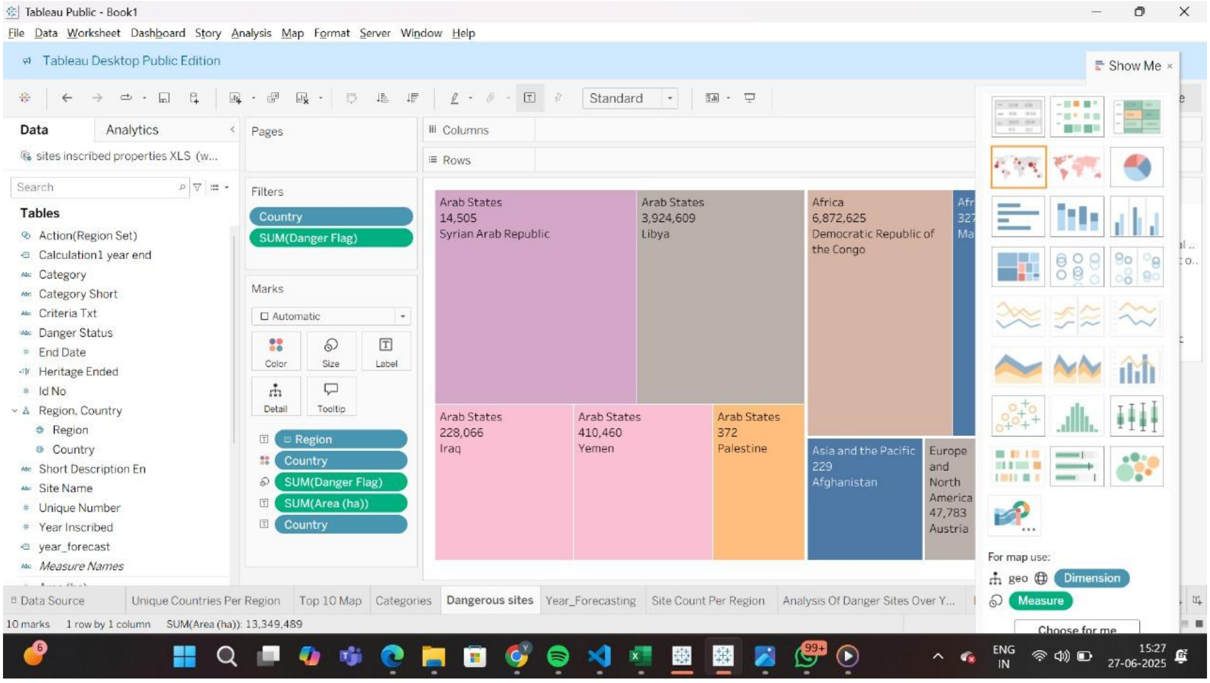


Tableau Public - Book1
Tableau Desktop Public Edition
Arab States
14,505
Syrian Arab Republic
Arab States
3,924,609
Libya
Africa
6,872,625
Democratic Republic of the Congo
Arab States
228,066
Iraq
Arab States
410,460
Yemen
Arab States
372
Palestine
Asia and the Pacific
229
Afghanistan
Europe and North America
47,783
Austria
Filters
Country
SUM(Danger Flag)
Marks
Region
Country
SUM(Danger Flag)
SUM(Area (ha))
Country
Data Source
Unique Countries Per Region
Top 10 Map
Categories
Dangerous sites
Year_Forecasting
Site Count Per Region
Analysis Of Danger Sites Over Y...
10 marks
1 row by 1 column
SUM(Area (ha)): 13,349,489

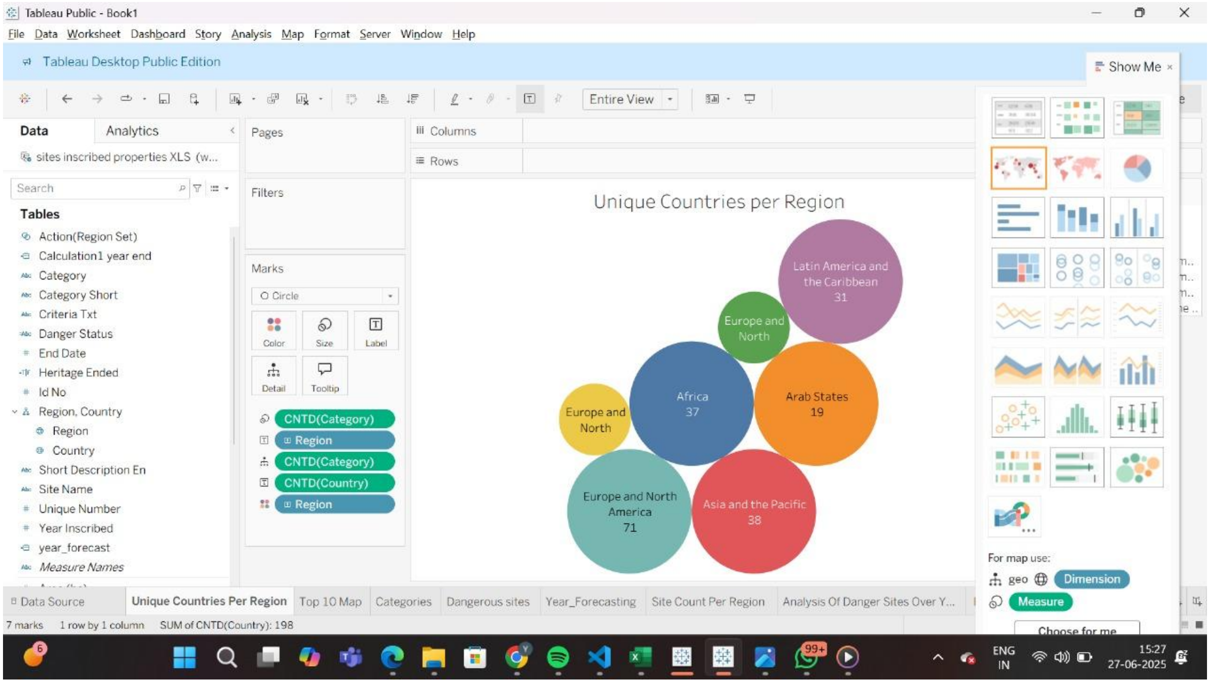


Tableau Public - Book1
Tableau Desktop Public Edition
Unique Countries per Region
Latin America and the Caribbean
31
Europe and North
Africa
37
Arab States
19
Europe and North
Europe and North America
71
Asia and the Pacific
38
Marks
CNTD(Category)
Region
CNTD(Category)
CNTD(Country)
Region
Data Source
Unique Countries Per Region
Top 10 Map
Categories
Dangerous sites
Year_Forecasting
Site Count Per Region
Analysis Of Danger Sites Over Y...
7 marks
1 row by 1 column
SUM of CNTD(Country): 198

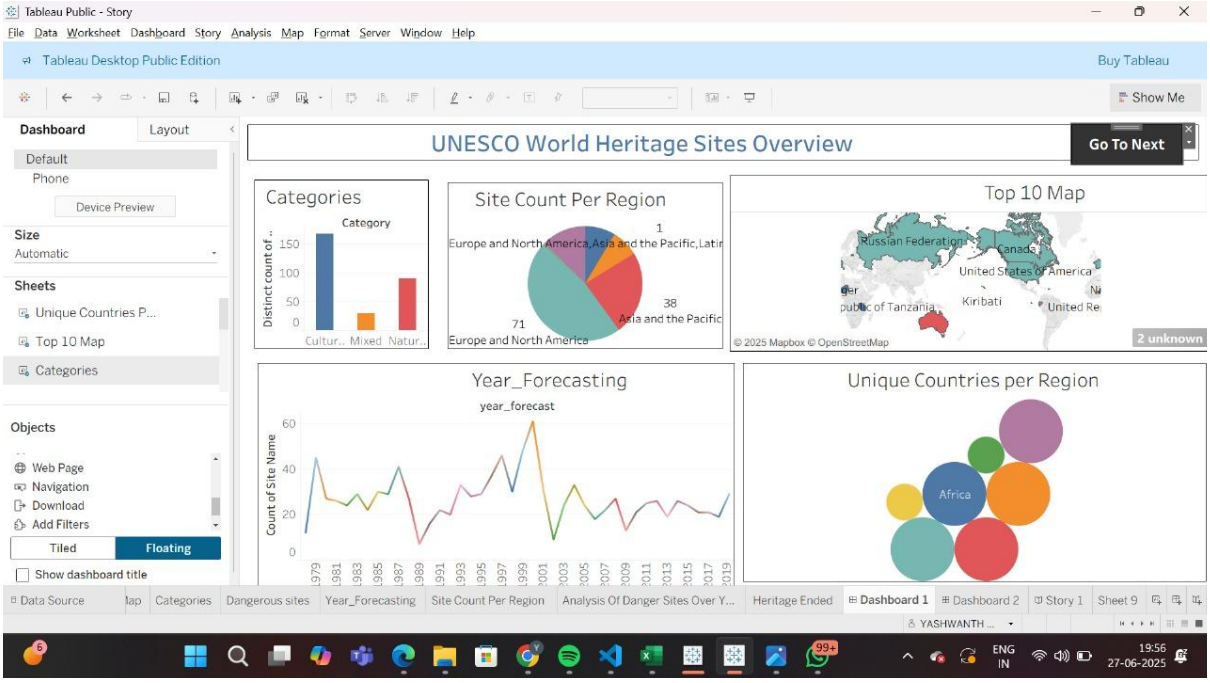


Tableau Public - Story
UNESCO World Heritage Sites Overview
Go To Next
Categories
Category
Distinct count of ..
150
100
50
0
Cultur..
Mixed
Natur..
Site Count Per Region
Europe and North America,Asia and the Pacific,Latir
1
38
Asia and the Pacific
71
Europe and North America
Top 10 Map
Russian Federation
Canada
United States of America
Kiribati
© 2025 Mapbox © OpenStreetMap
2 unknown
Year_Forecasting
year_forecast
Count of Site Name
60
40
20
0
Unique Countries per Region
Africa
Dashboard
Layout
Default
Phone
Device Preview
Size
Automatic
Sheets
Unique Countries P...
Top 10 Map
Categories
Objects
Web Page
Navigation
Download
Add Filters
Tiled
Floating
Show dashboard title
Data Source
Categories
Dangerous sites
Year_Forecasting
Site Count Per Region
Analysis Of Danger Sites Over Y...
Heritage Ended
Dashboard 1
Dashboard 2
Story 1
Sheet 9
19:56
27-06-2025

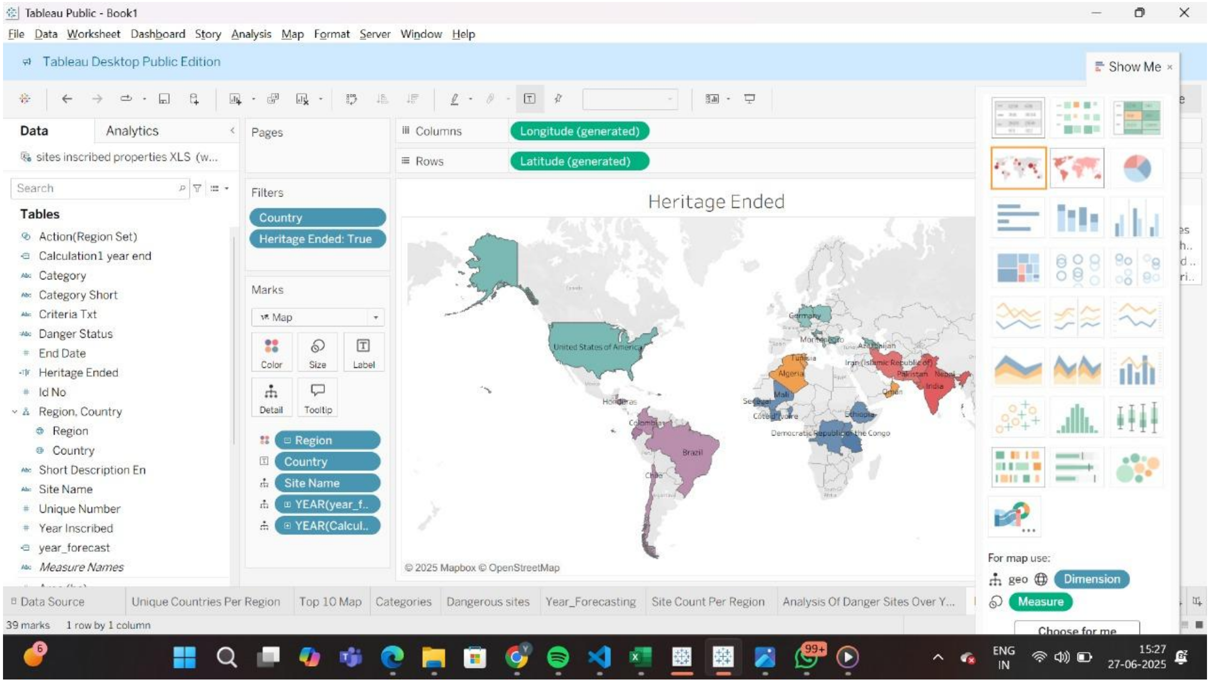


Tableau Public - Book1
Data
Analytics
sites inscribed properties XLS (w...
Tables
Action(Region Set)
Calculation1 year end
Category
Category Short
Criteria Txt
Danger Status
End Date
Heritage Ended
Id No
Region, Country
Region
Country
Short Description En
Site Name
Unique Number
Year Inscribed
year_forecast
Measure Names
Pages
Filters
Country
Heritage Ended: True
Marks
Map
Color
Size
Label
Detail
Tooltip
Region
Country
Site Name
YEAR(year_f..
YEAR(Calcul..
Columns
Longitude (generated)
Rows
Latitude (generated)
Heritage Ended
United States of America
Brazil
© 2025 Mapbox © OpenStreetMap
Show Me
For map use:
geo
Dimension
Measure
Data Source
Unique Countries Per Region
Top 10 Map
Categories
Dangerous sites
Year_Forecasting
Site Count Per Region
Analysis Of Danger Sites Over Y...
39 marks
1 row by 1 column
15:27
27-06-2025

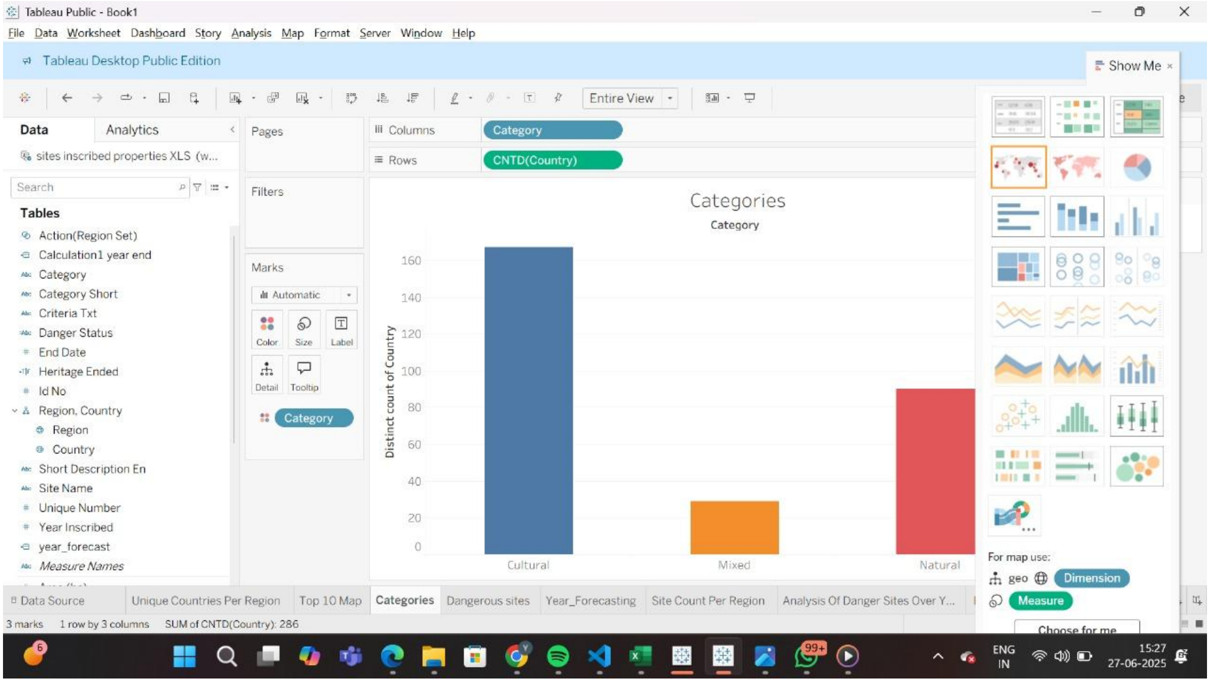


Tableau Public - Book1
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Tableau Desktop Public Edition
Show Me
Data Analytics
sites inscribed properties XLS (w...
Tables
Action(Region Set)
Calculation1 year end
Category
Category Short
Criteria Txt
Danger Status
End Date
Heritage Ended
Id No
Region, Country
Region
Country
Short Description En
Site Name
Unique Number
Year Inscribed
year_forecast
Measure Names
Pages
Filters
Marks
Automatic
Color Size Label
Detail Tooltip
Category
Columns Category
Rows CNTD(Country)
Categories
Category
Distinct count of Country
Cultural
Mixed
Natural
For map use:
geo Dimension
Measure
Data Source
Unique Countries Per Region
Top 10 Map
Categories
Dangerous sites
Year_Forecasting
Site Count Per Region
Analysis Of Danger Sites Over Y...
3 marks 1 row by 3 columns SUM of CNTD(Country): 286
15:27
27-06-2025

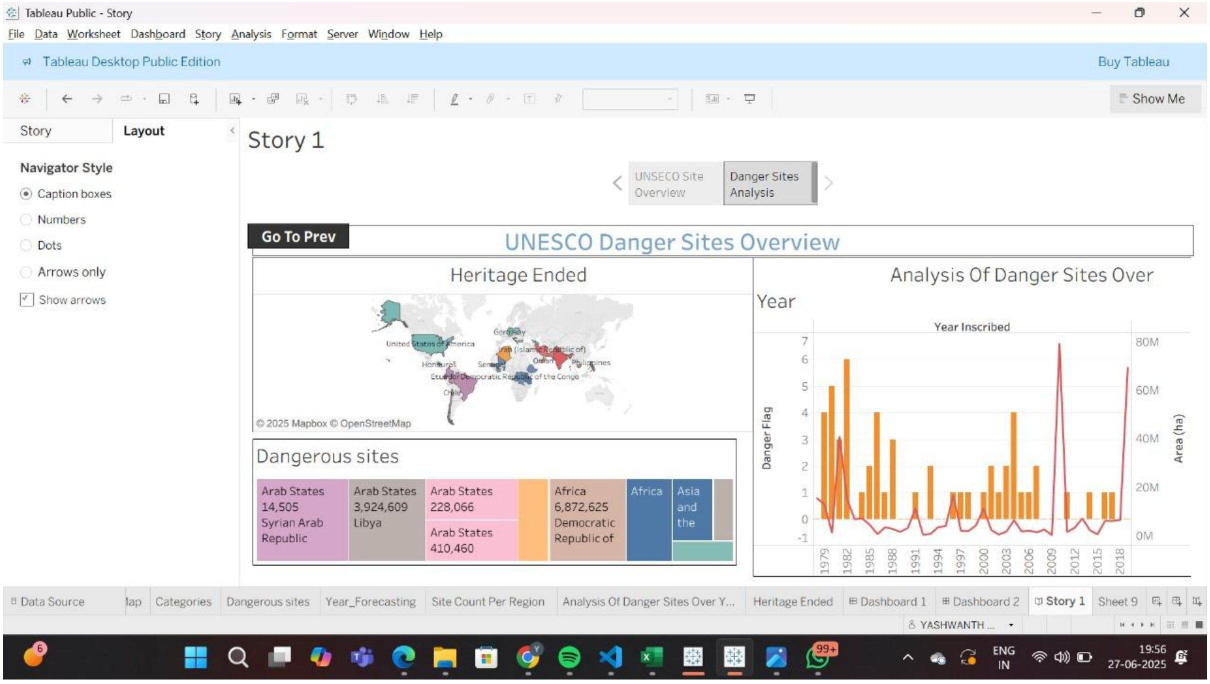
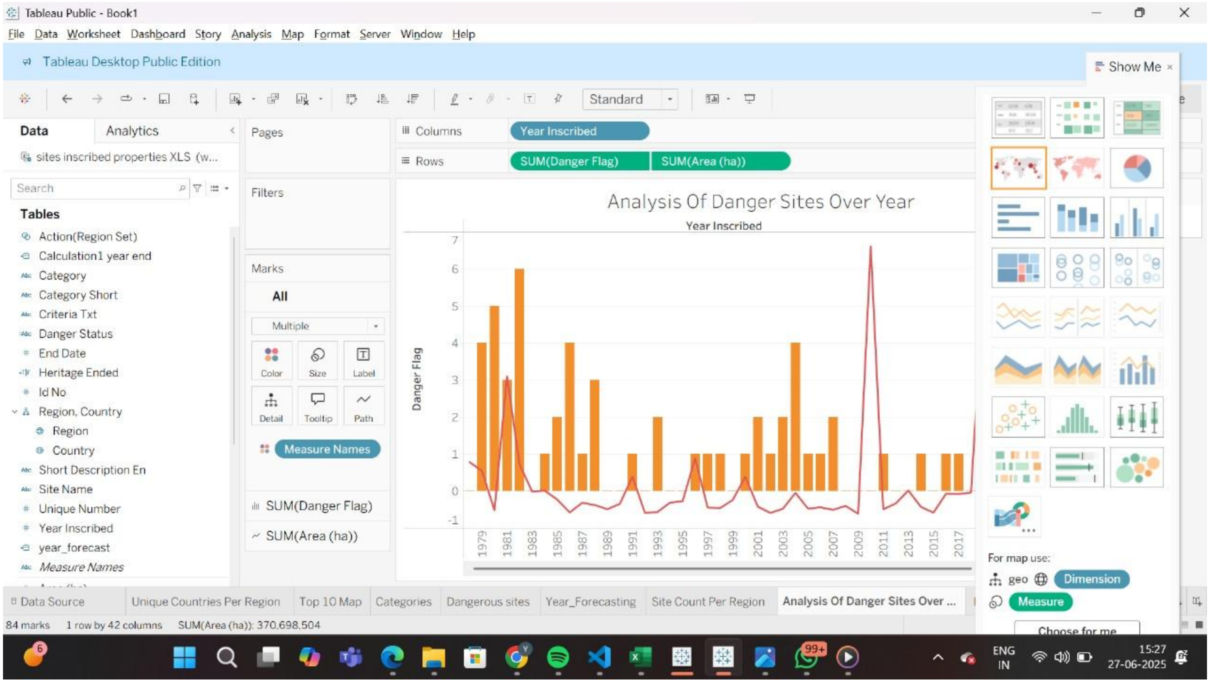
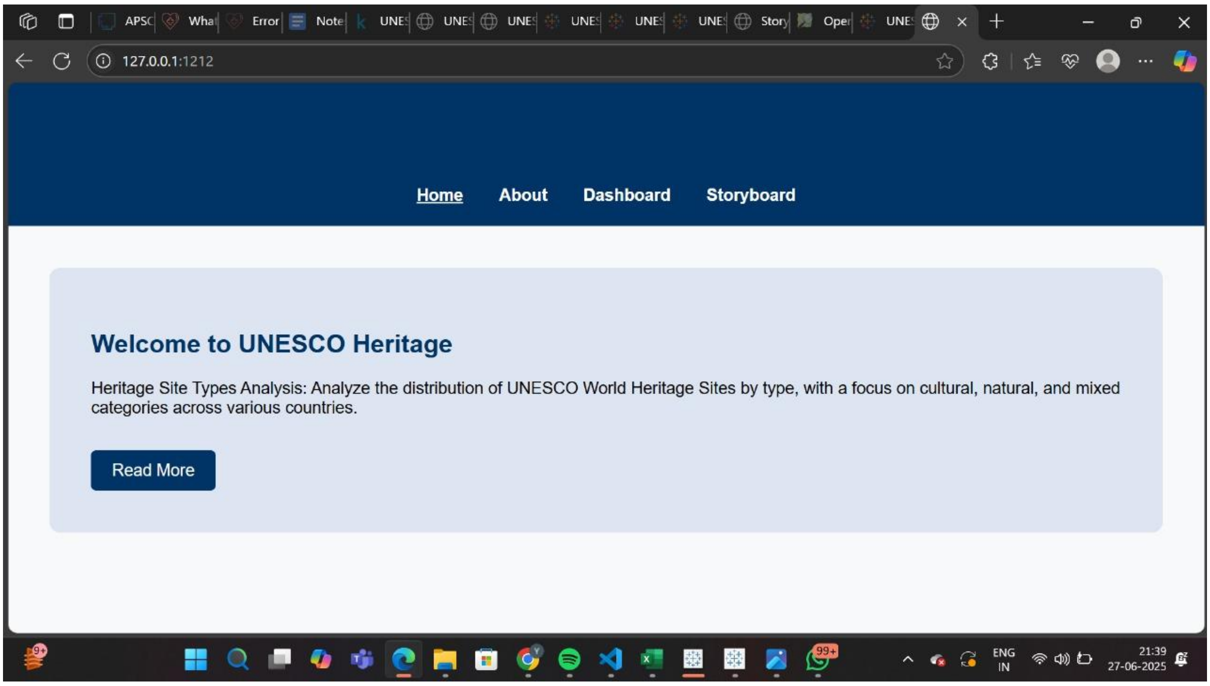


Tableau Public - Story
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Show Me
Story Layout
Navigator Style
Caption boxes
Numbers
Dots
Arrows only
Show arrows
Story 1
UNESCO Site Overview
Danger Sites Analysis
Go To Prev
UNESCO Danger Sites Overview
Heritage Ended
© 2025 Mapbox © OpenStreetMap
Dangerous sites
Arab States 14,505 Syrian Arab Republic
Arab States 3,924,609 Libya
Arab States 228,066
Arab States 410,460
Africa 6,872,625 Democratic Republic of
Africa
Asia and the
Analysis Of Danger Sites Over Year
Year Inscribed
Danger Flag
Area (ha)
Data Source
Categories
Dangerous sites
Year_Forecasting
Site Count Per Region
Analysis Of Danger Sites Over Y...
Heritage Ended
Dashboard 1
Dashboard 2
Story 1
Sheet 9
YASHWANTH ...
19:56
27-06-2025





8. ADVANTAGES & DISADVANTAGES

Advantages

- User-friendly, interactive dashboard
- Real-time filtering and drill-down
- Great visual understanding of global heritage diversity
- Promotes awareness and education

Disadvantages

- Tableau Public requires internet access
- Dataset may need frequent updates
- Some advanced filtering limited by Tableau constraints

---

9. CONCLUSION

This project successfully provides a dynamic platform to explore and understand UNESCO World Heritage Sites. By leveraging Tableau's capabilities, the data becomes more than just numbers—it becomes a story of human civilization and natural beauty worth preserving.

---

10. FUTURE SCOPE

- Integrate real-time updates from UNESCO API
- Add information on endangered sites and threats
- Enable storytelling with embedded videos or timelines
- Expand to include visitor statistics and tourism data
- Use AI for site recommendation or historical predictions